Prompt

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```
## [1] "276bfbde-9d13-4742-ab9f-724ee8cb4646"
```

Role

You are an expert academic assistant and an experienced instructor of R programming.

Task

Generate a comprehensive, plagiarism-free lecture slide for *Chapter N* based on the official textbook **R in a Nutshell: A Desktop Quick Reference** (2nd ed.), located at the repository root.

Instructions

1. **Structural Alignment**: Replicate the exact formatting, structural hierarchy, and stylistic layout of an existing chapter directory (e.g., `ch3/`) to ensure visual and pedagogical consistency across chapters. Produce both `chN.qmd` (English) and `chN-zh.qmd` (Traditional Chinese), structurally identical and cross-linked from the title slide.

(*Note*: Do not include section numbers like x.x or x.x.x in slide titles, except for the first page.)

2. **Exhaustive Coverage**: The handout must be meticulously detailed. Do not summarize or omit topics. Every code chunk, function, argument, and conceptual point introduced in the target chapter must be included and explained.

   (*Depth tiering — course-specific*: this exhaustive standard applies in full to Parts I–IV (Ch. 1–15), which each receive dedicated lecture time. Part V chapters (Ch. 16–23) are compressed into four teaching weeks and Part VI (Ch. 24–26) is self-study; for these, condensed coverage is acceptable — every *topic* must still appear, but secondary arguments and long option catalogs may be summarized, prioritizing runnable examples.)

3. **Paraphrasing & Academic Integrity**: To prevent plagiarism, completely rewrite and paraphrase all textual descriptions, explanations, and commentary from the textbook. The narrative flow and wording must be entirely original, academic, and tailored for a student audience, while strictly preserving the technical accuracy of the original content.

4. **Code Execution**: Ensure all R code syntax is updated, correct, and properly formatted in markdown code blocks. Chunks that install software or require unavailable services must use `eval=FALSE`.

5. **Dataset Loading Convention (course-specific)**: The textbook loads example data with `library(nutshell)` followed by `data(xxx)`. The `nutshell` package is no longer on CRAN, so **never** use it. Do **not** use `load(url(...))` either — it fails on some platforms. Instead, download each dataset from the CRAN GitHub mirror to a temporary file, then load it:

```r
temp <- tempfile(fileext = ".rda")
download.file("https://raw.githubusercontent.com/cran/nutshell/master/data/xxx.rda",
              temp, mode = "wb")
load(temp)
```

When a chunk loads several datasets, declare `temp` once and repeat the `download.file()` + `load()` pair. Datasets from the companion packages use the same pattern with `nutshell.bbdb` or `nutshell.audioscrobbler` in place of `nutshell`.

6. **Directory and Storage Architecture**: Replicate the established file structure utilized in the previous module. Dynamically generate and store the output within a newly designated directory (e.g., `chN/`), mirroring the organization and naming conventions of the existing chapter repositories (shared `bib.bib` and `logo.svg`).